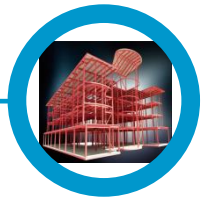




STEEL STRUCTURES/ SPACE STRUCTURES (CE-862) Fall Semester

Lecture 04


Beams



What are beams?

Although there are no hard-and-fast rules, the following names are in common use to describe beams with reference to their function:

Floor beam: In buildings, a major beam usually supporting joists; a transverse beam in bridge floors

Girder: In buildings, the same meaning as (and more commonly used than) floor beam; also any major beam in a structure

Girt: A horizontal member fastened to and spanning the wall columns of industrial buildings, used to support wall covering, such as corrugated metal

Header: A beam framed to two beams at right angles to it, and usually supporting joists on one side of it; used at openings such as stairwells

Joist: A beam supporting floor construction but not major beams

Lintel: A beam spanning door, window, or other wall openings and supporting wall immediately above

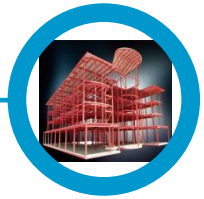
Purlin: A roof beam, usually supported by trusses

Rafter: A roof beam, usually supported by purlins

Spandrel beam: A beam at the outside wall of a building, supporting its share of the floor and also the wall up to the floor above

Stringer: In bridge floors, a longitudinal beam supported by floor beams (sometimes called a joist); a beam supporting stair steps

Trimmer: One of the beams or joists supporting a header



Sections used as beams?

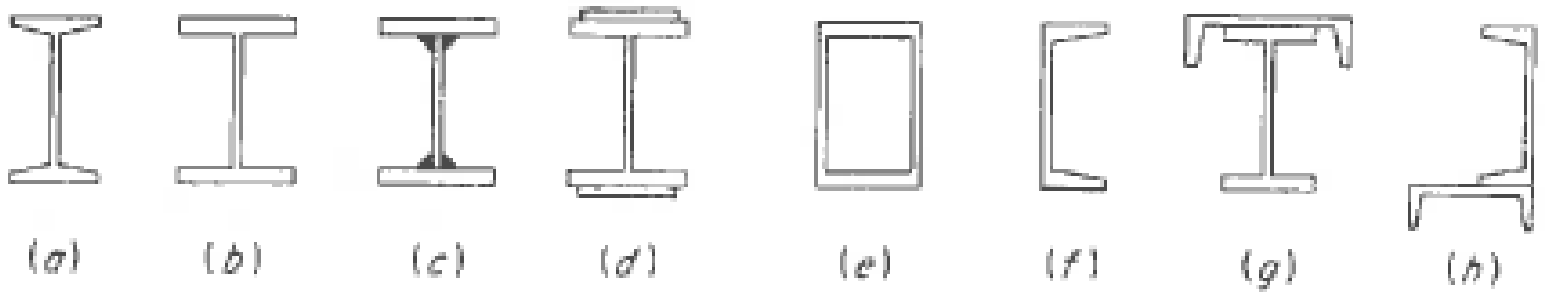
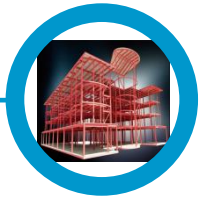


FIGURE 5-1



Bending Behaviour of beams?

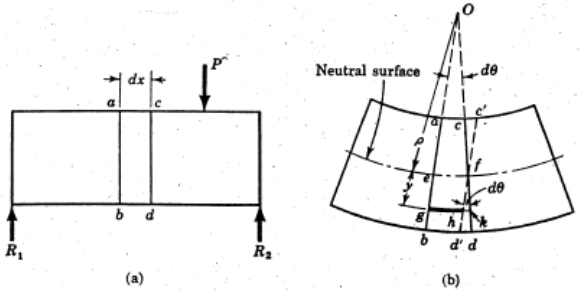


Figure 5-1 Deformations.

The plane containing fibers like *ef* is called the *neutral surface* because such fibers remain unchanged in length and hence carry no stress. It will be shown shortly that this neutral surface contains the centroids of all transverse sections.

Consider now the deformation of a typical fiber *gh* located *y* units from the neutral surface. Its elongation *hk* is the arc of a circle of radius *y* subtended by the angle $d\theta$ and is given by

$$\delta = hk = y\,d\theta$$

The strain is found by dividing the deformation by the original length *ef* of the fiber:

$$\epsilon = \frac{\delta}{L} = \frac{y\,d\theta}{ef}$$

If we denote the radius of curvature of the neutral surface by ρ, the curved length *ef* is equal to $\rho\,d\theta$; from which the strain becomes

$$\epsilon = \frac{y\,d\theta}{\rho\,d\theta} = \frac{y}{\rho}$$

Assuming that the material is homogeneous and obeys Hooke's law (assumption 2), the stress in fiber *gh* is given by

$$\sigma = E\epsilon = \left(\frac{E}{\rho}\right)y \qquad (a)$$

Equation (*a*) indicates that the stress in any fiber varies directly with its location *y* from the neutral surface, since it is assumed that the modulus of elasticity *E* is equal in tension and compression (assumption 3) and the radius of curvature ρ of the neutral surface is independent of the location *y* of the fiber. However, the stresses must not exceed the proportional limit, for this would invalidate Hooke's law on which this stress variation is based.

To complete the derivation of the flexure formula, we apply the conditions of equilibrium. As we saw in Art. 4-3, the external loads that act to one side of an exploratory section are balanced by the resisting shear V_r and the resisting moment M_r. To create this balance, a typical element in the exploratory section is subjected to the forces shown in the pictorial sketch* in Fig. 5-2. The line of intersection between the neutral surface and the transverse exploratory section is called the *neutral axis*, abbreviated NA.

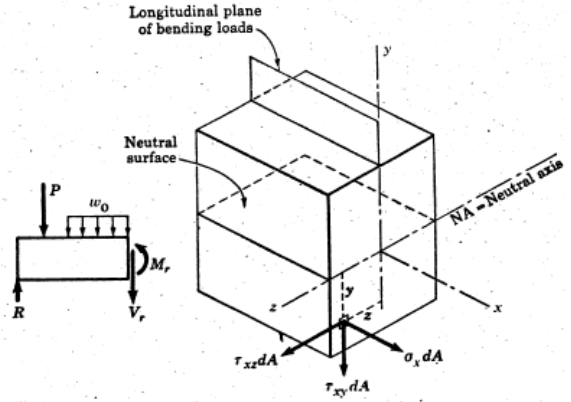


Figure 5-2 Forces acting on a typical element of the cross section of a beam.

To satisfy the conditions that the external loads have no *x* components (assumption 5), we must have

$$[\Sigma F_x = 0] \qquad \int \sigma_x\,dA = 0$$

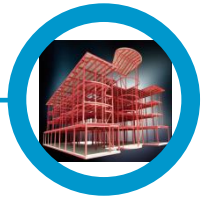
where σ_x is equivalent to σ in Eq. (*a*). On replacing σ_x by Ey/ρ, we obtain

$$\frac{E}{\rho}\int y\,dA = 0$$

The constant ratio E/ρ is written outside the integral sign. Since $y\,dA$ is the moment of the differential area dA about the neutral axis, the integral $\int y\,dA$ is the total moment of area. Hence

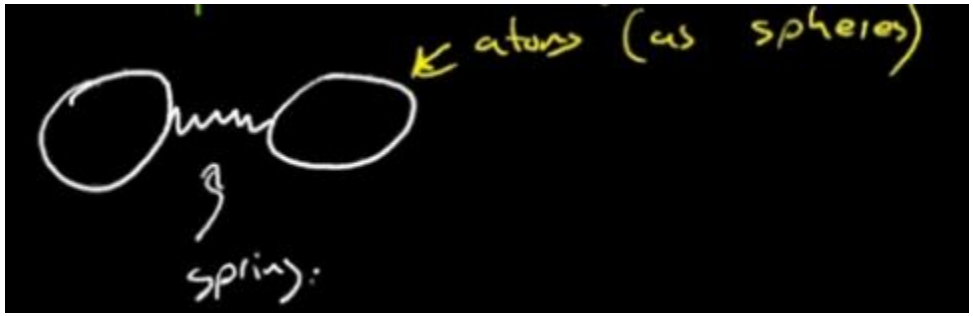
$$\frac{E}{\rho}A\bar{y} = 0$$

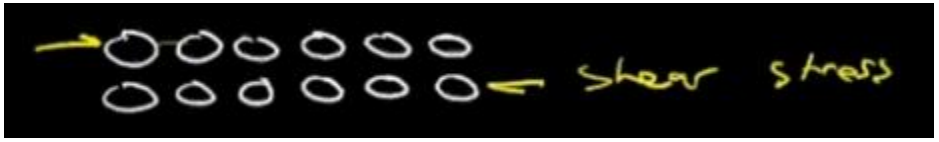
* The cross section is drawn as rectangular only for ease of representation; it may have any shape.

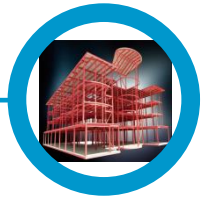


What is elastic and plastic deformation

Elastic: the sample when under loading return to original unloading
Plastic: sample don't return to original loading, atoms or molecules move past one another to new resting positions

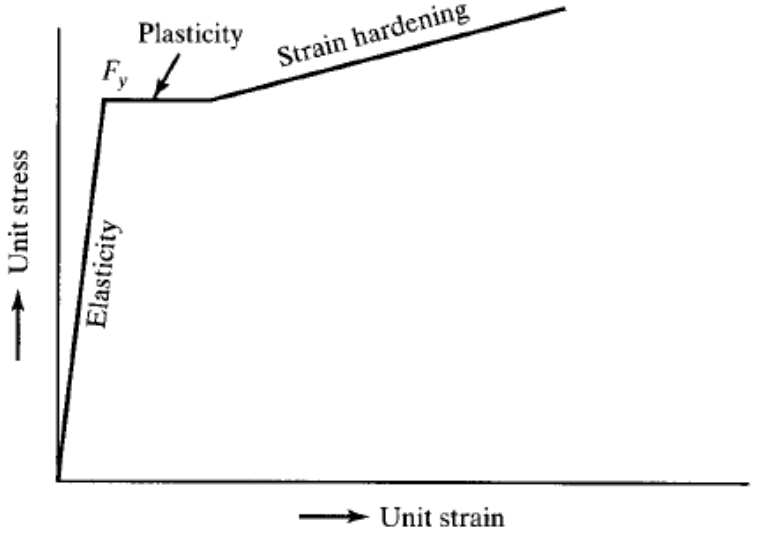


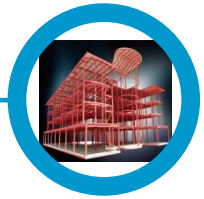




Theory of Plastic analysis

The basic plastic theory has been shown to be a major change in the distribution of stresses after the stresses at certain points in a structure reach the yield stress. The theory is that those parts of the structure that have been stressed to the yield stress cannot resist additional stresses. They instead will yield the amount required to permit the extra load or stresses to be transferred to other parts of the structure where the stresses are below the yield stress, and thus in the elastic range and able to resist increased stress. Plasticity can be said to serve the purpose of equalizing stresses in cases of overload.

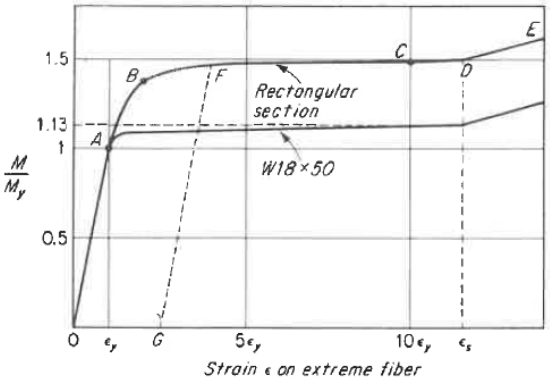




Bending Behaviour of beams?

As the load on the beam increases beyond the load corresponding to M_y strain continues to increase in proportion to distance from the neutral axis, bu the stress distribution consists of OA, the elastic region, and a portion of AB, th yield plateau, of Fig. 5-3, provided the extreme-fiber strain is less than ϵ_s, th strain at the beginning of strain hardening. Thus, if the extreme-fiber strain is $2\epsilon_y$ as in Fig. 5-4*d*, the stress distribution is that shown in Fig. 5-4*e*. The correspond ing resisting moment is

$$M = 2\left[\frac{bdF_y}{4}\left(\frac{d}{4}+\frac{1}{2}\times\frac{d}{4}\right)+\frac{bd}{4}\frac{F_y}{2}\frac{2}{3}\frac{d}{4}\right]=\frac{F_y bd}{4}\frac{3d}{4}+\frac{F_y bd}{8}\frac{d}{3}=\frac{11}{48}F_y bd^2 \quad (f)$$



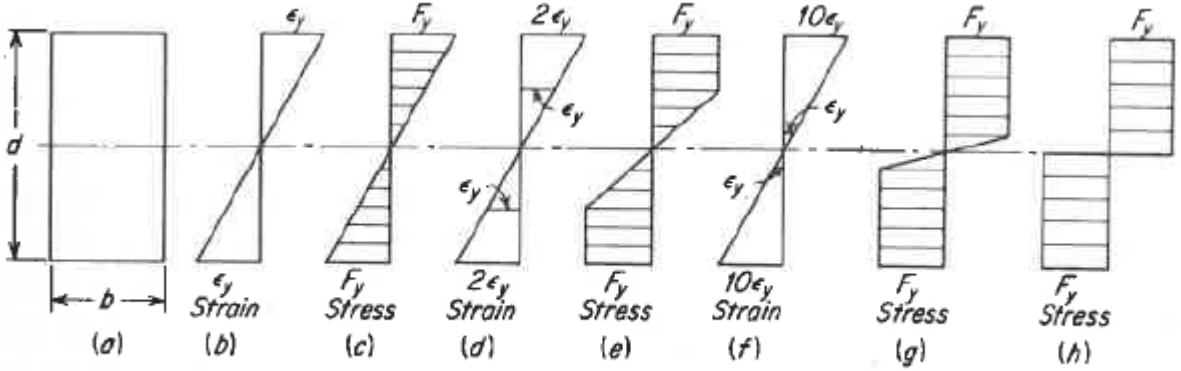
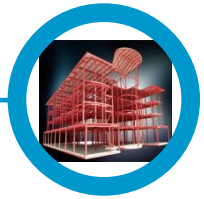
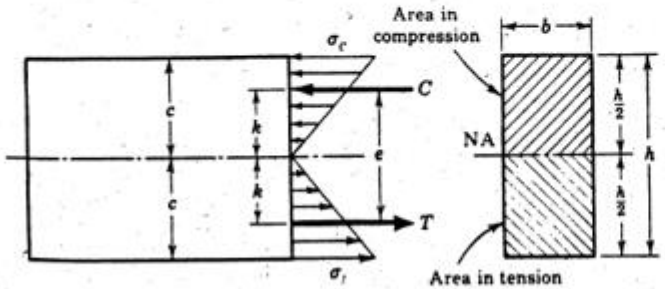


FIGURE 5-4



Bending Behaviour of beams?





$$T = C = (\sigma_{ave.})(\text{Area}) = \left(\frac{1}{2}\sigma\right)\left(b\,\frac{h}{2}\right)$$

$$M = M_r = Ce = Te$$

$$M = \left(\frac{1}{2}\sigma\right)\left(b\,\frac{h}{2}\right)\left(\frac{2}{3}h\right) = \sigma\,\frac{bh^2}{6}$$

The moment given by Eq. (*g*) is called the *plastic moment* of resistance, denoted by M_p. It is usually taken as the limiting value; i.e., the benefits of strain hardening are neglected. The ratio of the plastic moment to the yield moment for the rectangular cross section is given by

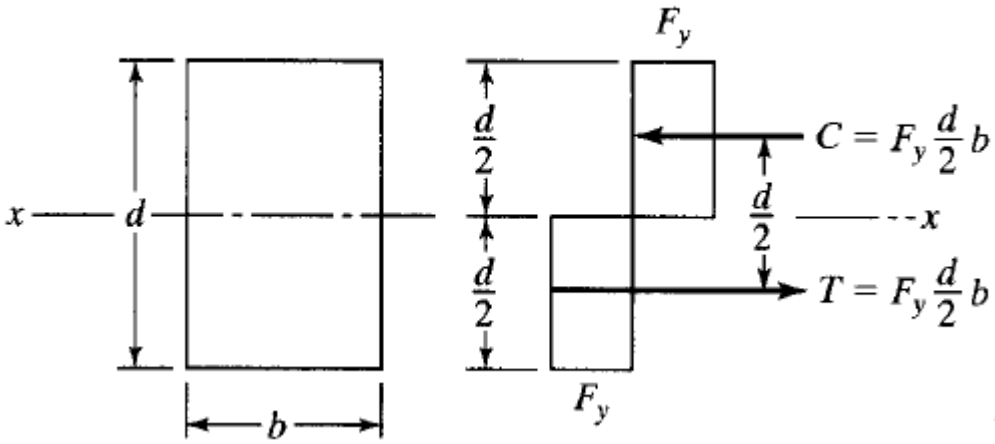
$$\frac{M_p}{M_y} = \frac{F_y bd^2/4}{F_y bd^2/6} = 1.5$$

This ratio is called the *shape factor*.

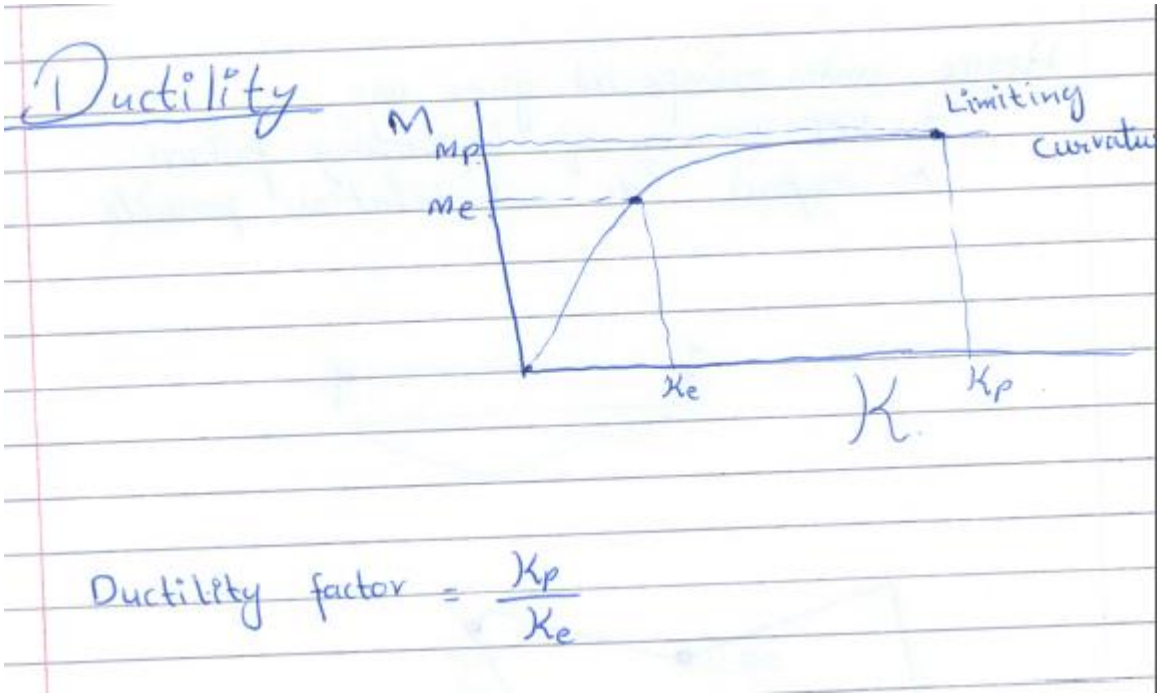
Circular	1.70
Thin-walled circular tube	1.27
Thin-walled rectangular tube	1.12



Bending Behaviour of beams?



Theory of Plastic analysis



The Mechanism method:

- Identify all possible collapse mechanisms.
- Work out the geometry of motion (kinematic) of each mechanism
- If a trial mechanism is a correct collapse mechanism then there is a neutral state in which energy balance of

Rate of work done by the loads = Rate of energy dissipated as heat in all plt

$$\dot{W} = \dot{D}$$

This tells us what value of a load factor is requ for collapse in trial mech

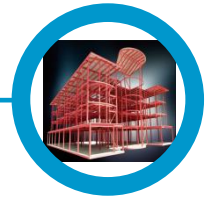
- Determine which trial mechanism is the correct collapse mech, & what is the value of load factor at collapse.

We could envisage searching through ${}^{c}C_{a-r}$ possible mechanisms

eg: $c = 4$ (critical sections) $\quad a = 2$ (static Indeterminacy)

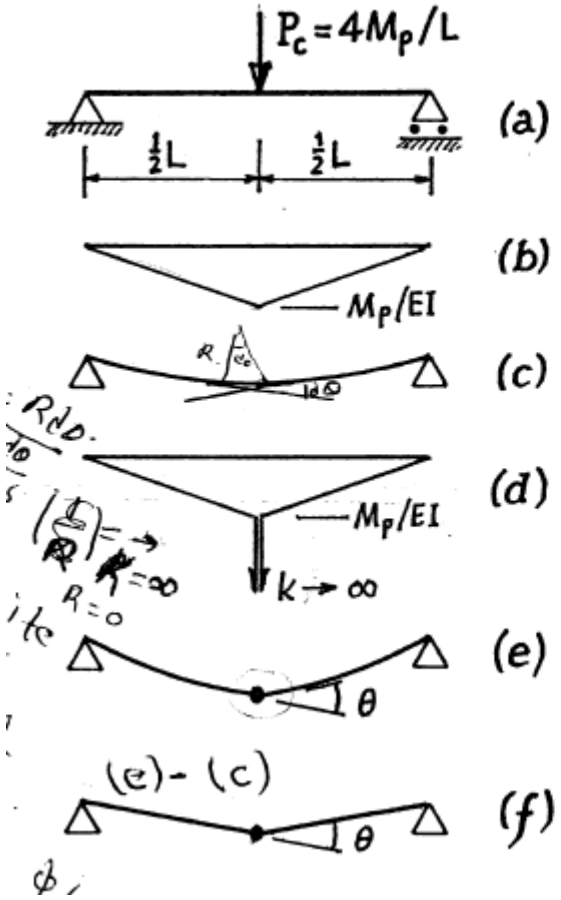
$${}^{4}C_{3} = \frac{4!}{(3)\,(1)} = 4$$

There appear to be 4 possible mechanisms in wh 3 plh are chosen from the 4 critical sectio



Theory of Plastic analysis

Simply supported beam at the state of plastic collapse



When the load attains the value $P_c = 4M_p/L$, Figure 2(a), all sections are elastic except for the central one where $M = M_p$. If this section is at point A or B on Figure 1(a), the curvature would be as in Figure 2(b) and the deflected configuration as in Figure 2(c), with no plastic deformation. However, if the central section is at point C, the curvature would be as in Figure 2(d) and the deflected configuration may now contain a plastic hinge rotation θ, Figure 2(e). In this example, θ is arbitrary and a mechanism is formed.

The incremental movement achieved *after* the load attains the value P_c is obtained by subtracting the deflected configuration of Figure 2(c) from that shown in Figure 2(e). The result shown in Figure 2(f) is called the *plastic collapse mechanism* and the load P_c at which it is activated is called the *plastic collapse load.*

Beams



Part I : The Principle of Virtual Displacements

Suppose that a beam or frame is in a displaced configuration in which the actual load $q(x)$ is in equilibrium with a bending moment distribution $M(x)$ and the static boundary conditions are satisfied. Then any virtual displacement $\bar{v}(x)$, which is fully compatible with a curvature distribution $\bar{k}(x)$ and which satisfies the kinematic boundary conditions, makes the external virtual work equal to the internal virtual work.

$$\int_s q(x)\bar{v}(x)dx = \int_s M(x)\bar{k}(x)dx$$

kinematic admissability
The structure doesn't break

▪ **Proof.** Since $M(x)$ is in equilibrium with $q(x)$, then $q(x) = -M''(x)$ (1)

Multiply both sides of (1) by $\bar{v}(x)$ and integrate over the whole beam or frame.

$$\int_s q(x)\bar{v}(x)dx = -\int_s M''(x)\bar{v}(x)dx \quad (2)$$

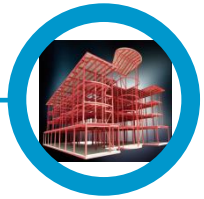
A twice-performed integration by parts converts (2) into (3).

$$\int_s q(x)\bar{v}(x)dx = [M(x)\bar{v}'(x) - M'(x)\bar{v}(x)]_s - \int_s M(x)\bar{v}''(x)dx \quad (3)$$

Since $\bar{v}(x)$ is compatible with $\bar{k}(x)$, then $\bar{k}(x) = -\bar{v}''(x)$ (4)

The boundary terms in (3) vanish if $M(x)$ and $\bar{v}(x)$ satisfy respectively any of the usual static and kinematic boundary conditions. Therefore, with (4), (3) becomes

$$\int_s q(x)\bar{v}(x)dx = \int_s M(x)\bar{k}(x)dx \quad (5)$$

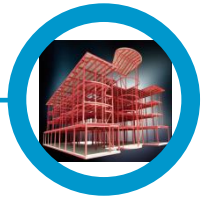


Virtual Work Method

The Standard Boundary Conditions

The actual and virtual systems are assumed to satisfy boundary conditions only from the standard set given in (11) to (14) below.

Simple Support	$M = 0$ (static),	$v = 0$ (kinematic)	(11a,b)
Fixed Support	$v = 0$ (kinematic),	$v' = 0$ (kinematic)	(12a,b)
Free End	$M = 0$ (static),	$V = M' = 0$ (static)	(13a,b)
Normal Slide	$v' = 0$ (kinematic),	$V = M' = 0$ (static)	(14a,b)



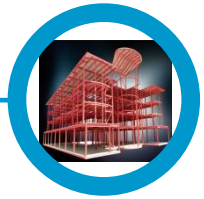
Virtual Work Method

Concentrated Loads and Deformations

Suppose that the actual static system contains concentrated loads W_i and moments C_i located at positions x_i in the beam or frame, as well as a continuous axial force distribution $N(x)$. Suppose also the actual kinematic system contains discrete rotational deformations θ_j located at x_j, as well as a continuous axial strain $\varepsilon(x)$. Then the integrals in the balance of external and internal virtual work can be extended to contain the additional discrete terms, quantities marked by an over-bar indicating corresponding virtual static or kinematic components.

$$\int_s q(x)\bar{v}(x)dx + \sum_i (W\bar{v} + C\bar{\phi})_i = \int_s [M(x)\bar{k}(x) + N(x)\bar{\varepsilon}(x)]dx + \sum_j M_j\bar{\theta}_j \qquad (17)$$

$$\int_s \bar{q}(x)v(x)dx + \sum_i (\bar{W}v + \bar{C}\phi)_i = \int_s [\bar{M}(x)k(x) + \bar{N}(x)\varepsilon(x)]dx + \sum_j \bar{M}_j\theta_j \qquad (18)$$



Theorems of plastic collapse

Static or lower bound theorem

If, at any load factor λ, it is possible to find a bending moment distribution in equilibrium with the applied loads and everywhere satisfying the yield condition, then λ is either equal to or less than the load factor at failure.

Kinematic or upper bound theorem

If, for any assumed plastic mechanism, the external work done by the loads at a positive load factor λ is equal to the internal work at the plastic hinges, then λ is either equal to or greater than the load factor at failure.

Beams



Consideration of these examples shows that a bending moment distribution which represents the collapse conditions must satisfy the following requirements.

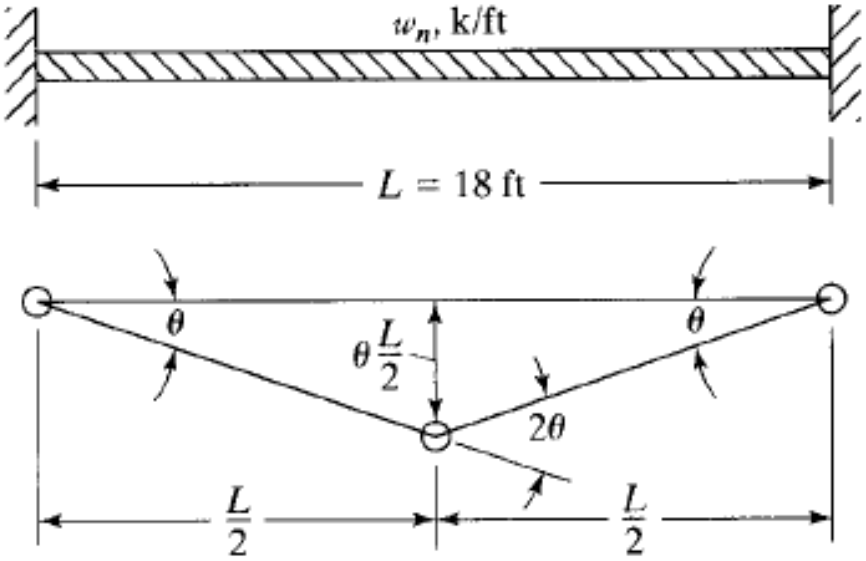
(a) *Equilibrium condition*. The bending moments must represent a state of equilibrium between the internal and external loads.

(b) *Mechanism condition*. The plastic moment of resistance must be reached at a sufficient number of sections, and in the necessary senses, for a collapse mechanism to form.

(c) *Yield condition*. The plastic moment of resistance (determined by the value of the yield stress) must nowhere be exceeded.



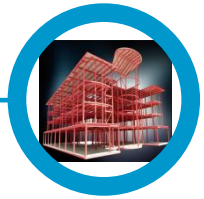
Virtual Work Method



$$M_n(\theta + 2\theta + \theta) = w_n L\left(\frac{1}{2} \times \theta \times \frac{L}{2}\right)$$

$$M_n = \frac{w_n L^2}{16}$$

$$w_n = \frac{16 M_n}{L^2}.$$



Virtual Work Method

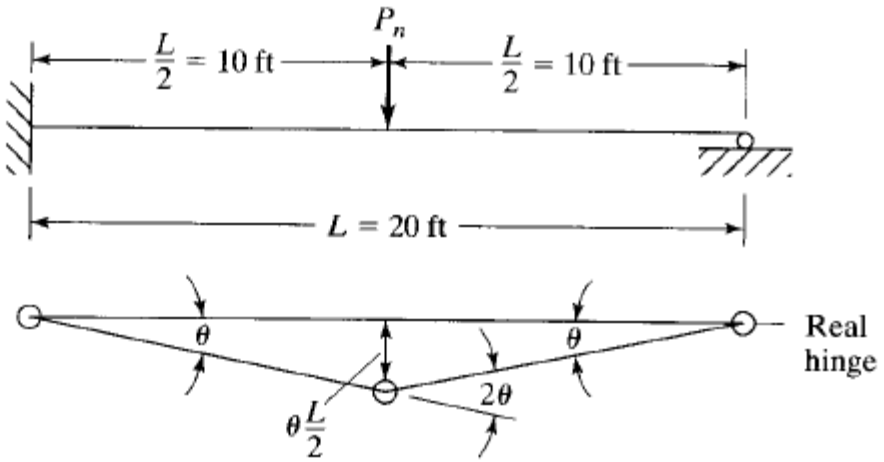


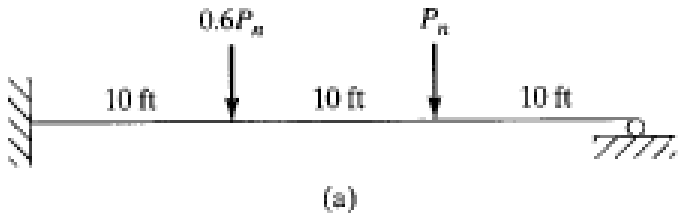
FIGURE 8.11

$$M_n(\theta + 2\theta) = P_n\left(\theta\frac{L}{2}\right)$$

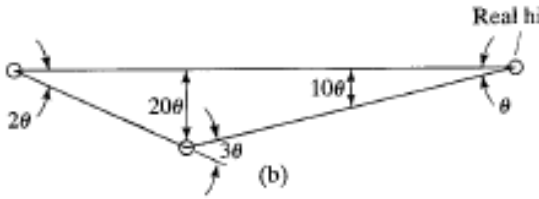
$$M_n = \frac{P_n L}{6} \quad (\text{or } 3.33P_n \text{ for the 20-ft beam shown})$$

$$P_n = \frac{6M_n}{L} \quad (\text{or } 0.3M_n \text{ for the 20-ft beam shown})$$

Beams

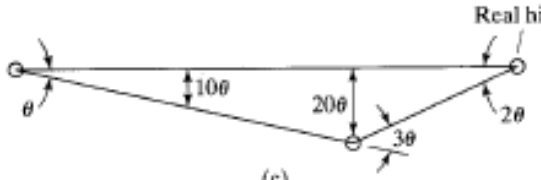


(a)



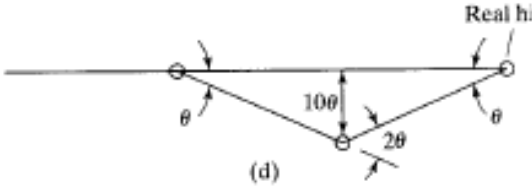
$M_n(5\theta) = (0.6P_n)(20\theta) + (P_n)(10\theta)$
$M_n = 4.4P_n$
$P_n = 0.227M_n$

(b)



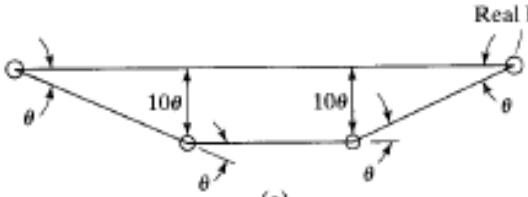
$M_n(4\theta) = (0.6P_n)(10\theta) + (P_n)(20\theta)$
$M_n = 6.5P_n$
$P_n = 0.154M_n$ ←

(c)



$M_n(3\theta) = (P_n)(10\theta)$
$M_n = 3.33P_n$
$P_n = 0.3M_n$

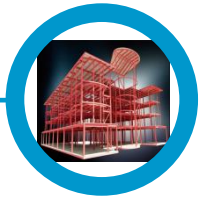
(d)

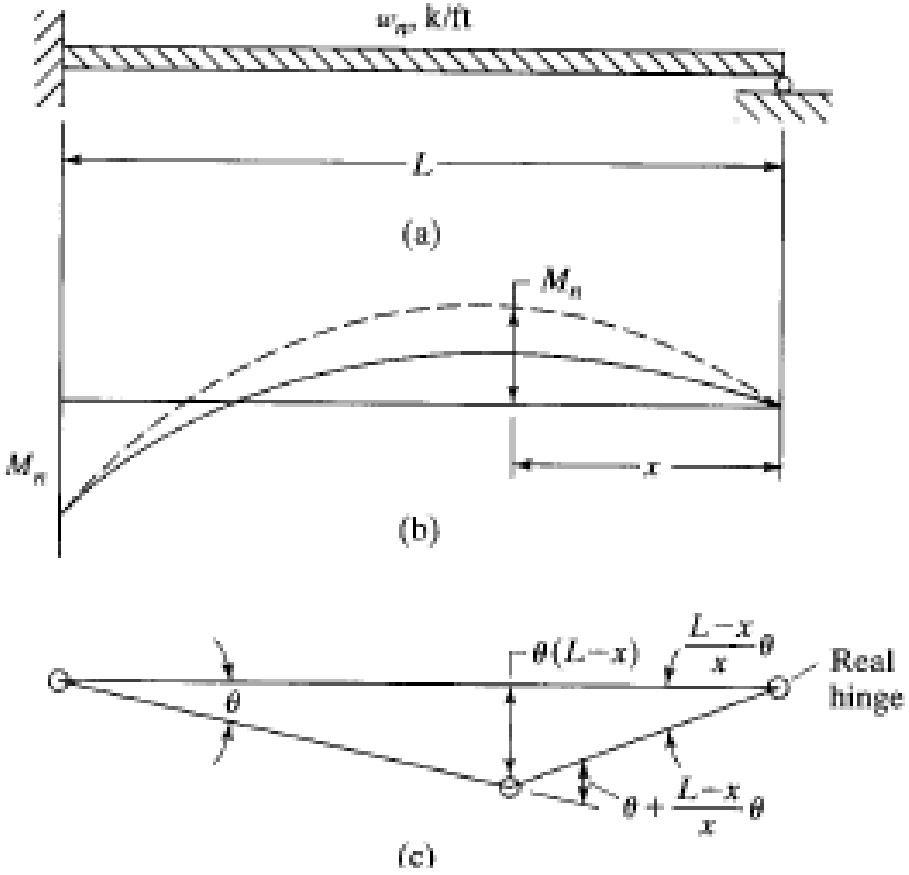


$M_n(3\theta) = (0.6P_n)(10\theta) + (P_n)(10\theta)$
$M_n = 5.33P_n$
$P_n = 0.1875M_n$

(e)

Beams





$$M_n\left(\theta + \theta + \frac{L-x}{x}\theta\right) = (w_n L)(\theta)(L-x)\left(\frac{1}{2}\right).$$

External work done = intensity of udl x area of plastic deformation(i.e. area of triangle ½ pxb

Solving this equation for M_n, taking $dM_n/dx = 0$, the value of x can be calculated to equal $0.414L$. This value is also applicable to uniformly loaded end spans of continuous beams with simple end supports, as will be illustrated in the next section.

Beams

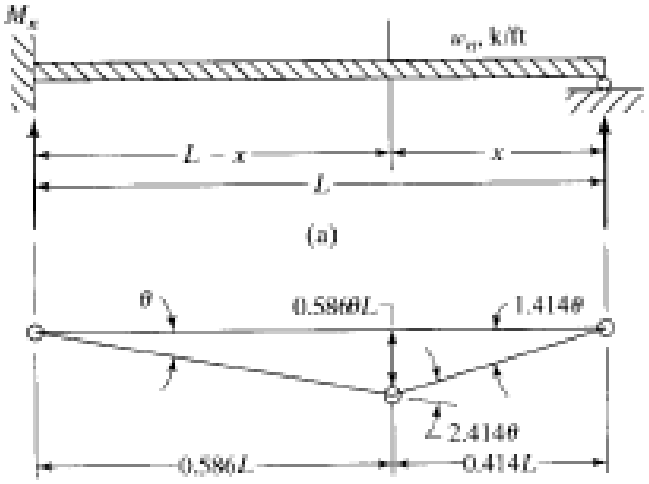


FIGURE 8.15

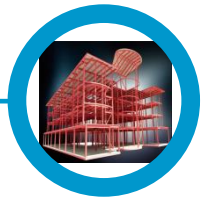
The beam and its collapse mechanism are redrawn in Fig. 8.15, and the following expression for the plastic moment and uniform load are written by the virtual-work procedure:

$$M_n(\theta + 2.414\theta) = (w_n L)(0.586\theta L)\left(\frac{1}{2}\right)$$

$$M_n = 0.0858 w_n L^2$$

$$w_n = 11.65\frac{M_n}{L^2}$$

Beams



A W18 × 55 ($Z_x = 112\ \text{in}^3$) has been selected for the beam shown in Fig. 8.16. Using 50 ksi steel and assuming full lateral support, determine the value of w_n.

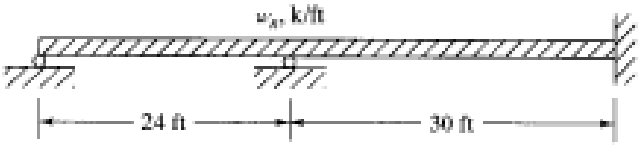
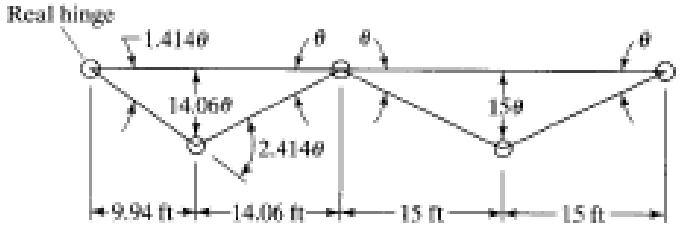


FIGURE 8.16

Solution

$$M_n = F_y Z = \frac{(50\ \text{ksi})(112\ \text{in}^3)}{12\ \text{in/ft}} = 466.7\ \text{ft-k}$$

Drawing the (collapse) mechanisms for the two spans:



Left-hand span:

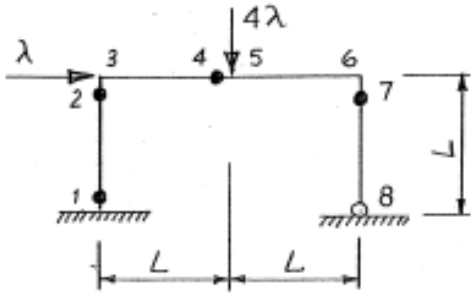
$$(M_n)(3.414\theta) = (24w_n)\left(\frac{1}{2}\right)(14.06\theta)$$

$$w_n = 0.0202\ M_n = (0.0202)(466.7) = 9.43\ \text{k/ft}$$

Right-hand span:

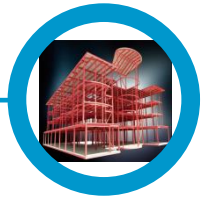
$$(M_n)(4\theta) = (30w_n)\left(\frac{1}{2}\right)(15\theta)$$

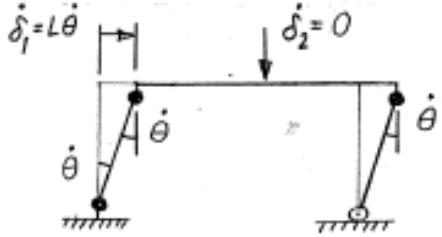
$$w_n = 0.0178\ M_n = (0.0178)(466.7) = 8.31\ \text{k/ft} \leftarrow$$



Columns 12, 78 : M_p

Beams 34, 56 : $2M_p$



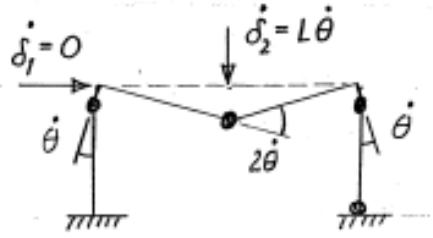


MECHANISM 1

$\dot{W} = \lambda(L\dot{\theta}) + 4\lambda(0) = \lambda L\dot{\theta}$

$\dot{D} = M_p\dot{\theta} + M_p\dot{\theta} + M_p\dot{\theta} = 3M_p\dot{\theta}$

$\dot{W} = \dot{D} \rightarrow \lambda = 3M_p/L$

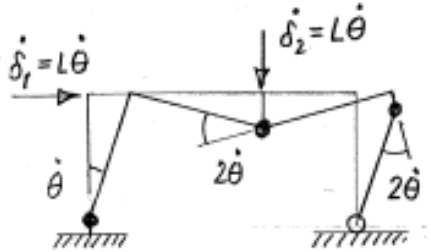


MECHANISM 2

$\dot{W} = \lambda(0) + 4\lambda(L\dot{\theta}) = 4\lambda L\dot{\theta}$

$\dot{D} = M_p\dot{\theta} + 2M_p(2\dot{\theta}) + M_p\dot{\theta} = 6M_p\dot{\theta}$

$\dot{W} = \dot{D} \rightarrow \lambda = 1{\cdot}5\ M_p/L$

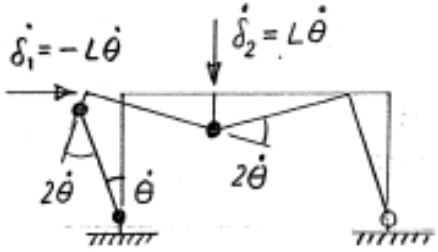


MECHANISM 3

$\dot{W} = \lambda(L\dot{\theta}) + 4\lambda(L\dot{\theta}) = 5\lambda L\dot{\theta}$

$\dot{D} = M_p\dot{\theta} + 2M_p(2\dot{\theta}) + M_p(2\dot{\theta}) = 7M_p\dot{\theta}$

$\dot{W} = \dot{D} \rightarrow \lambda = 1{\cdot}4\ M_p/L$



MECHANISM 4

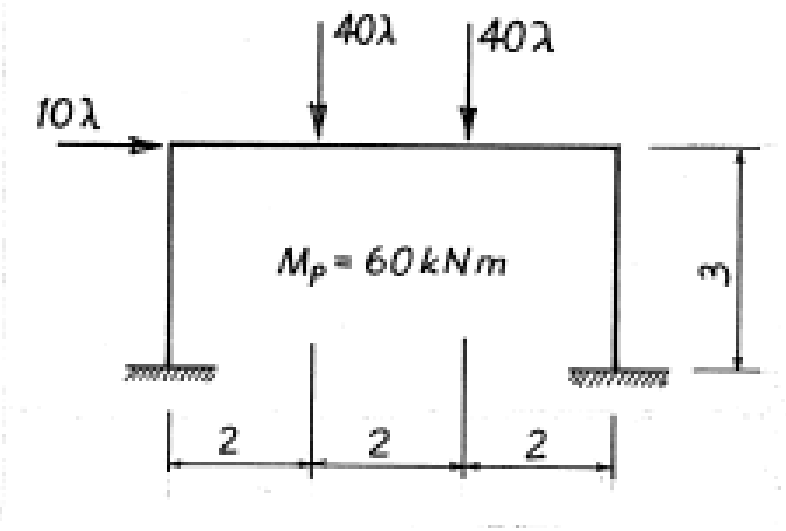
$\dot{W} = \lambda(-L\dot{\theta}) + 4\lambda(L\dot{\theta}) = 3\lambda L\dot{\theta}$

$\dot{D} = M_p(2\dot{\theta}) + M_p(2\dot{\theta}) + 2M_p(2\dot{\theta}) = 7M_p\dot{\theta}$

$\dot{W} = \dot{D} \rightarrow \lambda = 2{\cdot}33\ M_p/L$

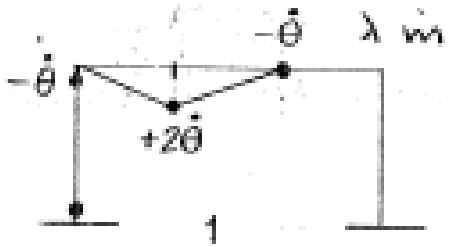
$\lambda_c = \text{Min}\ \lambda = 1.4\ M_p/L$

Beams

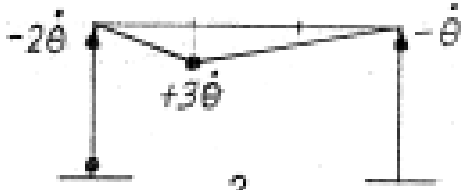


MECHANISMS IN A PORTAL FRAME $n=6 \quad \alpha=3$

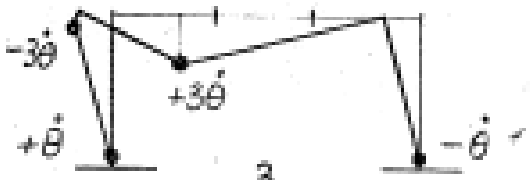
$^{n}C_{\alpha+1}$



1
$\lambda = 3.00$

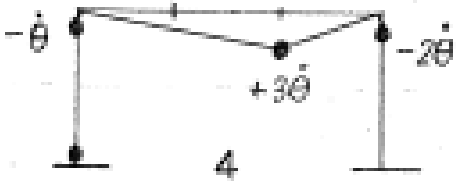


2
$\lambda = 1.50$
MINIMUM

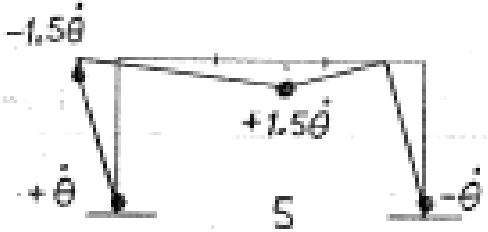


3
$\lambda = 2.29$

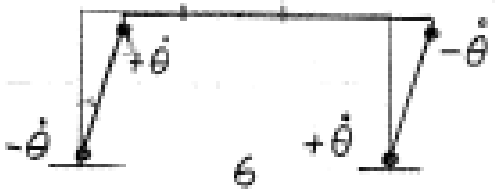
$^{6}C_{4} = 6!/(4!\,2!) = 15$



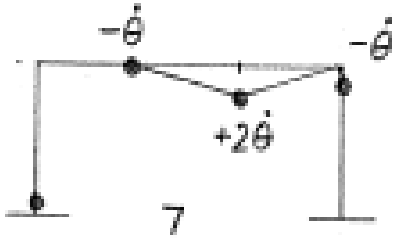
4
$\lambda = 1.50$
MINIMUM



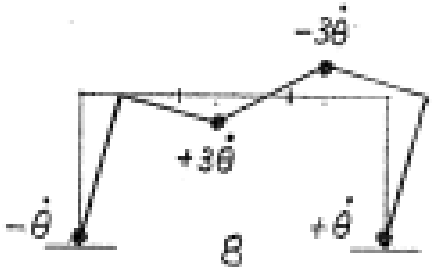
5
$\lambda = 3.33$



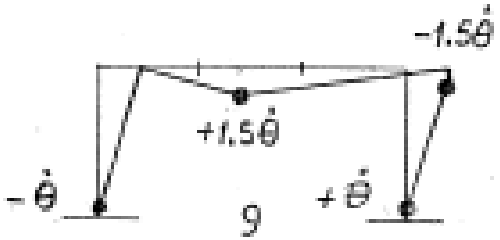
6
$\lambda = 8$



7
$\lambda = 3.00$

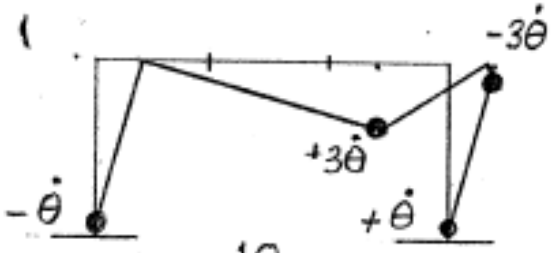


8
$\lambda = 16.00$



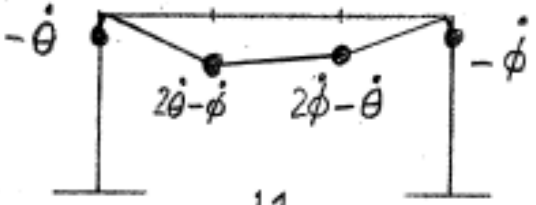
9
$\lambda = 2.00$

Beams



10

$\lambda = 1.78$



11

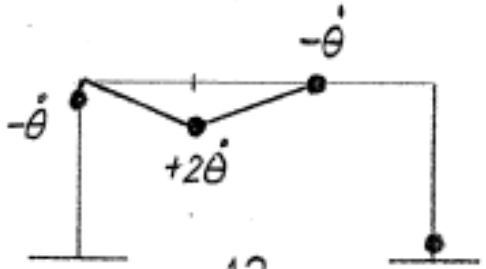
$\lambda = 1.50$

MINIMUM

Same as 2 + 4 2DOF

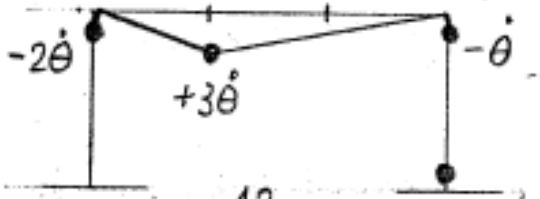
or 1 + 7 2DOF

Note that this is 2 DOF because we have 0 in p



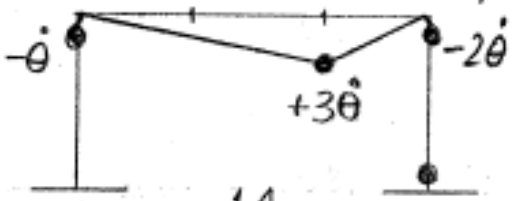
12

same as 1



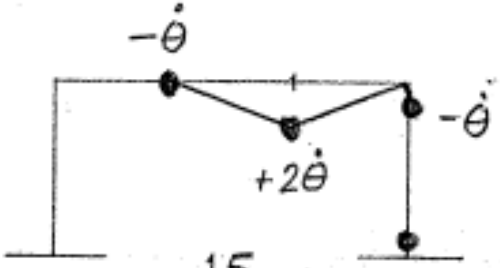
13

same as 2



14

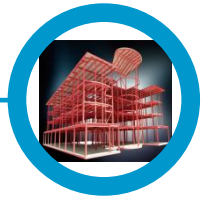
same as 4



15

same as 7

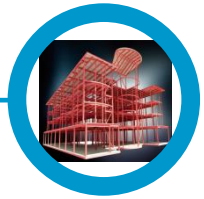
10 different 1 DOF mechanisms, 1 2DOF mechanism



General Notes

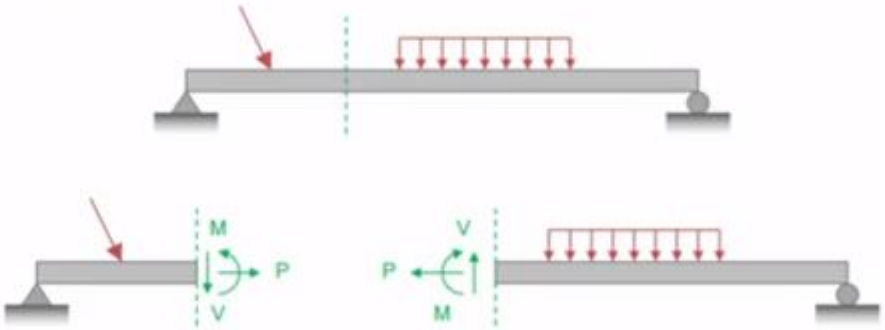
- Like columns, beams present a stability issue.
  - Under bending, a portion of the beam's cross-section is in compression.
- Also, like columns, a beam's capacity is a function of its unbraced length.
  - What we will call "L_b"
- However, a beam may reach its full capacity before "beam buckling" can occur.

- Let's begin with a discussion of bending stresses in general.



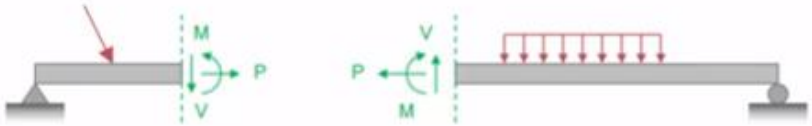
General Notes

- Recall that when we cut a section in a beam, we can generate at most three unknown components:

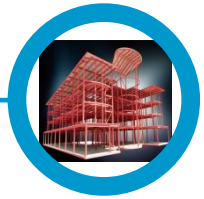


$$\sum F_x = 0$$
$$\sum F_y = 0$$
$$\sum M_z = 0$$

  - Note the 3 components from the 3 eqns. of equilibrium!

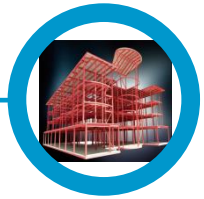


  - Axial forces (P) generate normal stresses (i.e. $\sigma = P/A$)
  - Bending moments (M) also generate normal stresses (i.e. σ = ???)
  - Shear forces (V) generate shear stresses (i.e. τ = ???)
    - We'll handle these later.



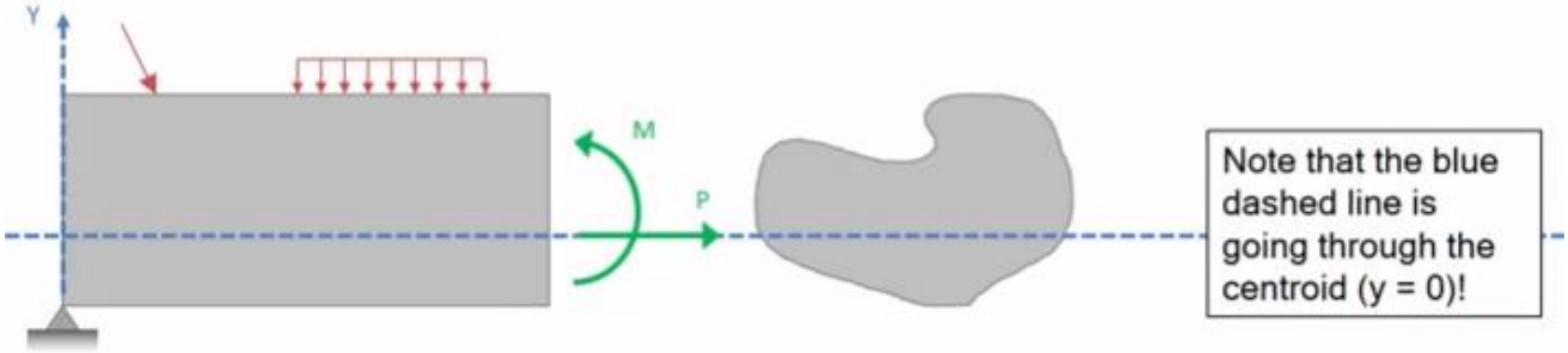
General Notes

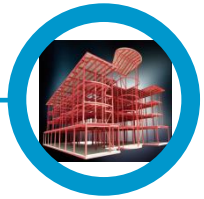
- To derive the normal stresses in beams, we'll assume the following:
  - Linear and elastic behavior:
    - Material follows Hooke's law: $\sigma = E\,\varepsilon$
  - Strain-compatibility:
    - That is, plane sections remain plane
    - In other words, all of the deformation is linearly proportional
  - Small deflections (i.e. first-order behavior):
    - That is, we don't need to consider equilibrium of the displaced section.
    - This will make more sense later when we discuss beam deflection.



General Notes

- If we use these assumptions, we can treat the problem as follows:
  - We have a beam of an arbitrary cross-section subjected to loads.
  - Using statics, we can determine "P" and "M" inside the beam.
  - We know that "P" and "M" will cause normal stress!
    - Question is, though, how much?

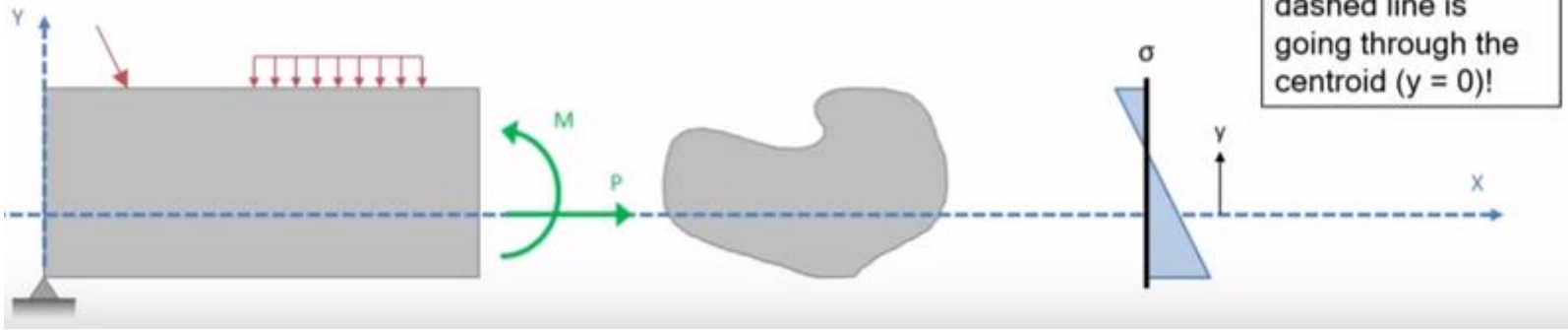




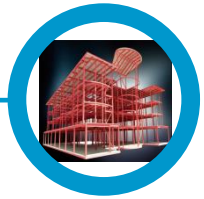
General Notes

- Remember, we're treating the beam's response as linear:
  - Therefore we can assume that the resulting stress on the beam is linear:
  - Therefore, as a function of "y," we can assume:

$$\sigma = c_1 + c_2 y$$



Note that the blue dashed line is going through the centroid (y = 0)!



General Notes

- After solving, for a symmetric section, we get the following:

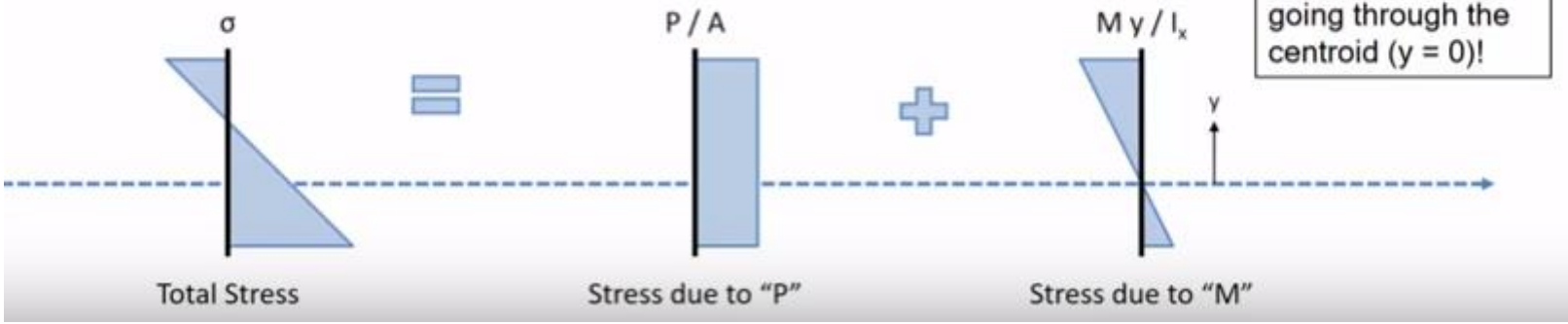
$$\sigma = c_1 + c_2 y$$

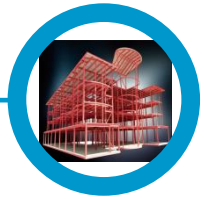
$$c_1 = P/A$$

$$c_2 = M/I_x$$

$$\sigma = \frac{P}{A} + \frac{My}{I_x}$$

Note that the blue dashed line is going through the centroid (y = 0)!





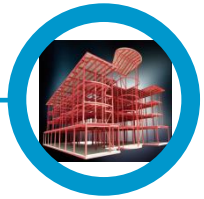
General Notes

- So, the stresses in a doubly-symmetric section due to general bending are:

$$\sigma(y) = \frac{P}{A} + \frac{My}{I_x}$$

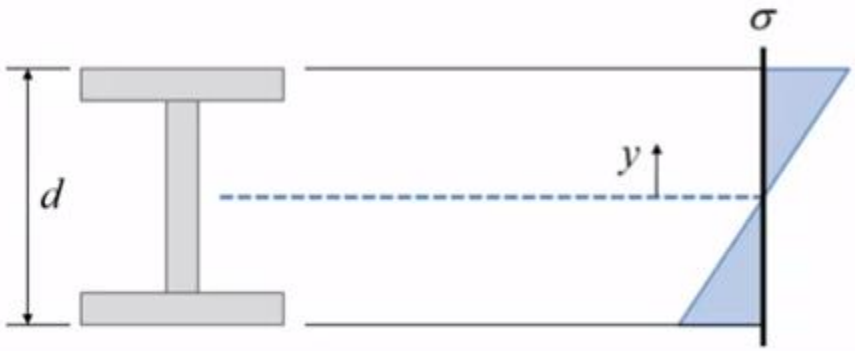
  - Let's consider a section with just strong-axis moment applied.

$$\sigma(y) = \frac{My}{I_x}$$



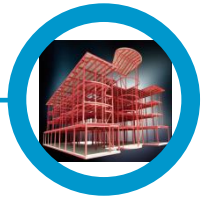
General Notes

- From this relationship, the maximum stresses in this section from bending are at the extreme fibers, or at d/2.
  - We define the denominator as the section modulus, S_x:



$$\sigma\left(\frac{d}{2}\right) = \frac{M}{I_x}\left(\frac{d}{2}\right) = \frac{M}{\left(\frac{I_x}{d/2}\right)}$$

$$\therefore \boxed{S_x = \frac{I_x}{d/2}}$$

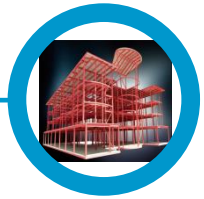


General Notes

- Because the section is doubly-symmetric, there is only one section modulus.
- If the section were singly-symmetric, we would need to define two section moduli:

$$S_{xc} = \frac{I_x}{c_{xc}} \qquad S_{xt} = \frac{I_x}{c_{xt}}$$

  - S_{xc} = section modulus to the compression flange
    - c_{xc} = distance from the centroid to the extreme compression fiber
  - S_{xt} = section modulus to the tension flange
    - c_{xt} = distance from the centroid to the extreme tension fiber

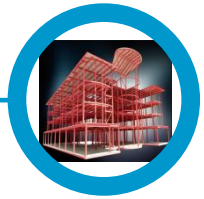


General Notes

- This relationship will be linear until the extreme fiber stresses reach F_y.
  - Once this occurs, this relationship will be no longer valid.
  - We call this point the yield moment, or M_y.

$$\sigma\left(d/2\right) = F_y = \frac{M}{S_x}$$

$$\therefore \boxed{M_y = F_y S_x}$$



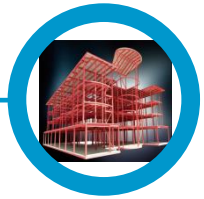
General Notes

- Like before, if the section in question is doubly-symmetric, there is only one yield moment.

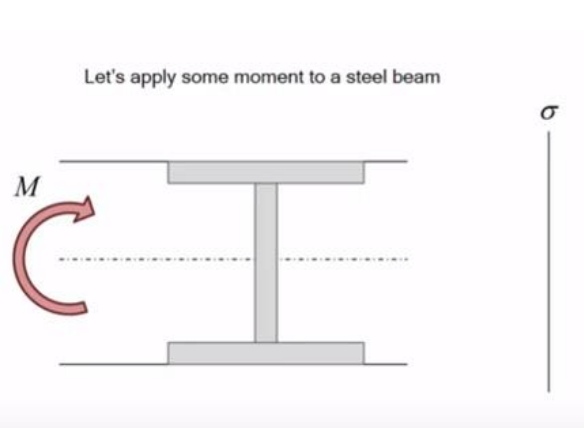
- If the section in question were singly-symmetric, we would need to define two yield moments:
  - M_{yc} = yield moment with respect to the compression flange
  - M_{yt} = yield moment with respect to the tension flange
  - The lower of these two values would govern.

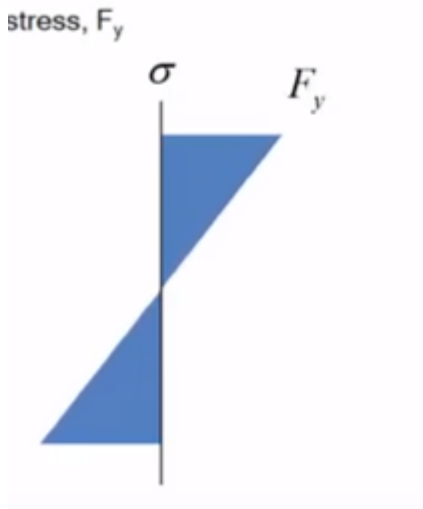
- Question:
  - Is the yield moment the ultimate capacity of a steel beam?

- Answer:
  - No!
  - As we'll see, steel beams will be able to withstand moments much larger than M_y.
    - This, however, is true only under certain conditions, which we'll discuss.



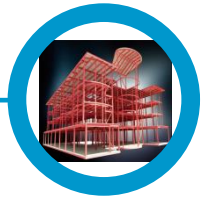
General Notes





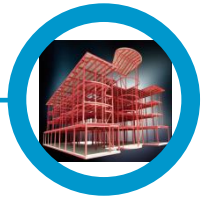


- Therefore, the plastic moment, or M_p, is the largest moment that a steel beam can take.
  - It is the point at which the entire cross-section has reached F_y.
  - After this point, the beam will freely deform under no increase in load.
- Therefore, we need a means of computing this value.



General Notes

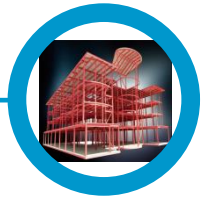
- Recall:
  - The elastic neutral axis of a cross-section is the location of the centroid.
- We'll now introduce a new definition.
  - The plastic neutral axis is where the compressive forces equal the tensile forces.
    - This, of course, has to be true no matter what.
    - However, we will relate it to the forces present in the section at the plastic moment.
- Aren't these the same thing?
  - Not necessarily.
  - Because of symmetry, the plastic neutral axis might not be in the same location as the neutral axis.



General Notes

- Now, consider a singly-symmetric section:
  - E.N.A. and P.N.A. have shifted to maintain C=T.





General Notes

- Remember this?
  - The E.N.A. and the P.N.A. aren't in the same place!

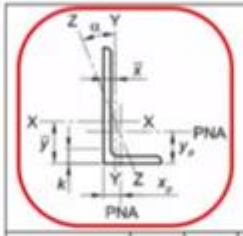
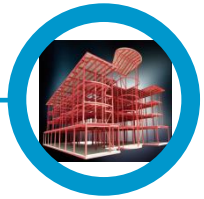


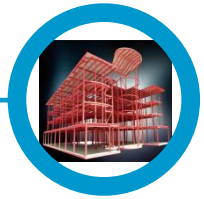
Table 1-7 (continued)
Angles
Properties

Shape	k	Wt.	Area, A	Axis X-X						Flexural-Torsional Properties		
				I	S	r	$\bar{y}$	Z	y_p	J	C_w	$\bar{r}_o$
	in.	lb/ft	in.^{2}	in.^{4}	in.^{3}	in.	in.	in.^{3}	in.	in.^{4}	in.^{6}	in.
L5×3½×	11/16	12.8	3.75	9.43	2.89	1.58	1.74	5.12	1.25	0.322	0.444	2.38
×7/16	7/8	11.3	3.31	8.41	2.56	1.59	1.72	4.53	1.22	0.220	0.304	2.39
×3/8	13/16	9.80	2.86	7.35	2.22	1.60	1.69	3.93	1.19	0.141	0.196	2.41
×5/16	3/4	8.20	2.41	6.24	1.87	1.61	1.67	3.32	1.14	0.0832	0.116	2.42
×1/4	11/16	6.60	1.94	5.09	1.51	1.62	1.64	2.68	1.12	0.0438	0.0606	2.43



General Notes

- There are three primary chapters of the AISC Specifications related to beams:
  - Chapter F:
    - Design of Members for Flexure
  - Chapter G:
    - Design of Members for Shear (we'll discuss this later on)
    - For now, we'll simply use a design aid.
  - Chapter L:
    - Design for Serviceability (i.e. deflections)



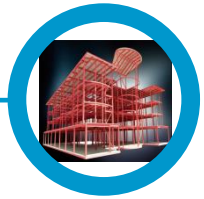
General Notes

- Once we determine where the P.N.A. is, we can determine M_p
- We do this by summing the moments of the plastic forces about the P.N.A.
  - The plastic forces are just the yield stress multiplied by the areas in compression and areas in tension.

- Therefore:

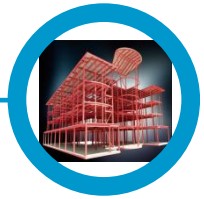
$$\begin{aligned} M_p &= P_C y_C + P_T y_T \\ &= F_y A_C y_C + F_y A_T y_T \\ &= F_y \left(A_C y_C + A_T y_T \right) \end{aligned}$$

  - This quantity in the parentheses defines a new section property.
  - We call this the plastic section modulus, Z_x.



General Notes

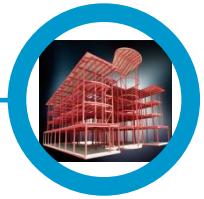
- Note:
  - Z_x is only a meaningful quantity if every component of the cross-section has the same F_y.
    - What's called a homogeneous section
  - If we have, say a plate girder, where different steels are used for the flanges and web, we just sum the moments of the forces directly.
    - This is called a hybrid section
    - Commonly used in bridge applications



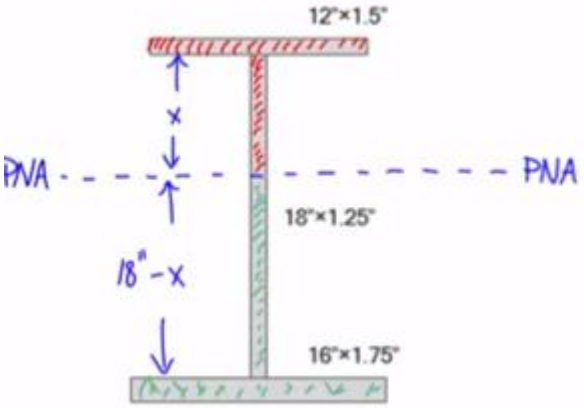
General Notes

- Let's compute Z_x for the following section:
  - For this section we'll compute the plastic moment capacity, M_p, as well.
    - All steel in the cross-section has F_y = 50 ksi.





General Notes



$A_c = (12)(1.5) + 1.25x$

$= 18 + 1.25x$

$A_T = (16)(1.75) + 1.25(18-x)$

$= 28 + 22.5 - 1.25x$

$= 50.5 - 1.25x$

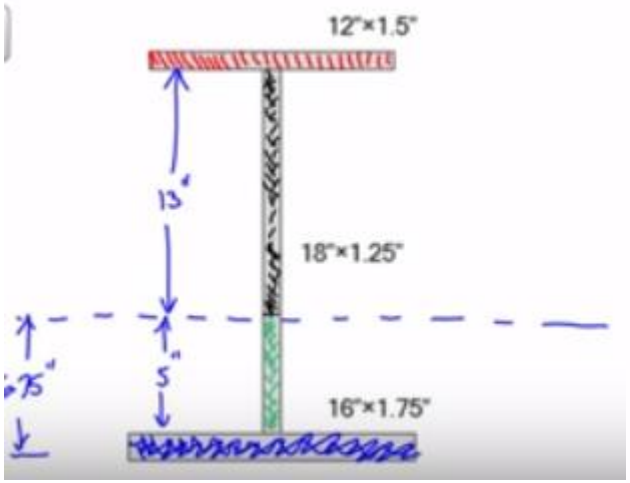
$\therefore \quad A_c = A_T$

$18 + 1.25x = 50.5 - 1.25x$

$-18 \quad +1.25x \quad -18 \quad +1.25x$

$2.5x = 32.5$

$x = 13$



$Z_x = (12)(1.5)\left(13 + \frac{1.5}{2}\right)$

$+ (13)(1.25)\left(\frac{13}{2}\right)$

$+ (5)(1.25)\left(\frac{5}{2}\right)$

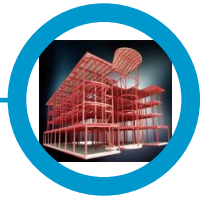
$+ (16)(1.75)\left(5 + \frac{1.75}{2}\right)$

$\therefore \boxed{Z_x = 533.25 \text{ in}^3}$

ANSWERS

$\therefore \quad M_p = F_y Z_x = (50 \text{ ksi})(533.25 \text{ in}^3) = 26662.5 \text{ in-kip}$

$\boxed{M_p = 2221.9 \text{ ft-kip}}$

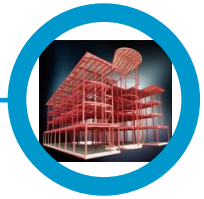


Lateral Bracing

- A beam, like a column, presents a stability issue.
  - Beams, like columns, are susceptible to buckling.
    - Flange and a portion of the web are in compression.
  - While these two types of buckling are different, both are a function of the length of the beam between bracing elements.
  - We denote this the unbraced length, or L_b.

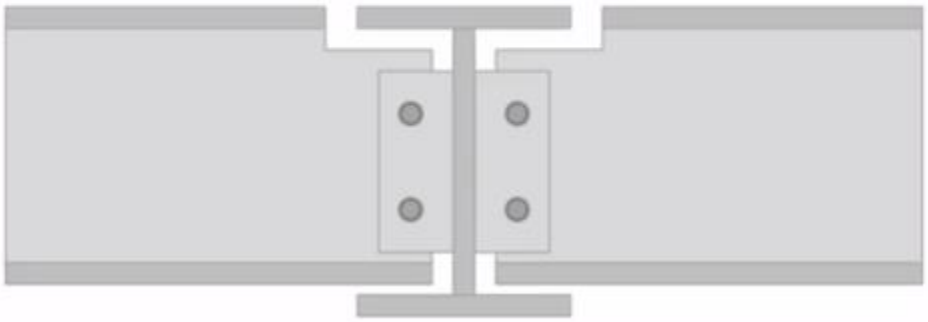
- Examples of lateral bracing:
  - Continuous support from floor slab (i.e. $L_b = 0$):



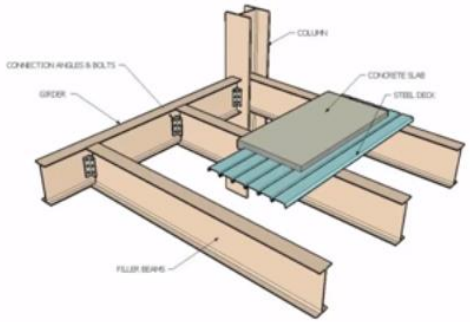


Lateral Bracing

- Examples of lateral bracing:
  - Beam framing in (on one or both sides):



- Example of a typical steel building floor system:

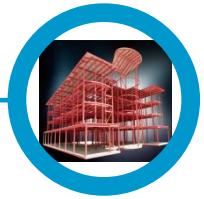


Lateral Bracing

- Example of cross frames in a steel girder bridge:

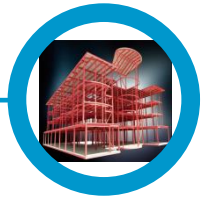






Lateral Bracing

- There are three primary chapters of the AISC Specifications related to beams:
  - Chapter F:
    - Design of Members for Flexure
  - Chapter G:
    - Design of Members for Shear (we'll discuss this later on)
    - For now, we'll simply use a design aid.
  - Chapter L:
    - Design for Serviceability (i.e. deflections)



Lateral Bracing

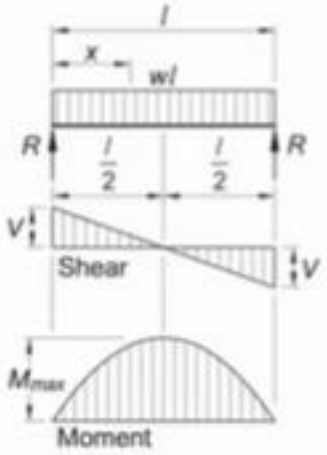
Page-3-215

- Table 3-23 is a very useful aid:

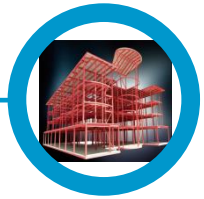
Table 3-23

Shears, Moments and Deflections

1. SIMPLE BEAM — UNIFORMLY DISTRIBUTED LOAD

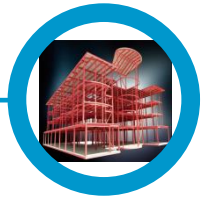


Total Equiv. Uniform Load		$= wl$
$R = V$		$= \frac{wl}{2}$
V_x		$= w\left(\frac{l}{2} - x\right)$
M_{max}	(at center)	$= \frac{wl^2}{8}$
M_x		$= \frac{wx}{2}(l - x)$
Δ_{max}	(at center)	$= \frac{5wl^4}{384EI}$
Δ_x		$= \frac{wx}{24EI}\left(l^3 - 2lx^2 + x^3\right)$



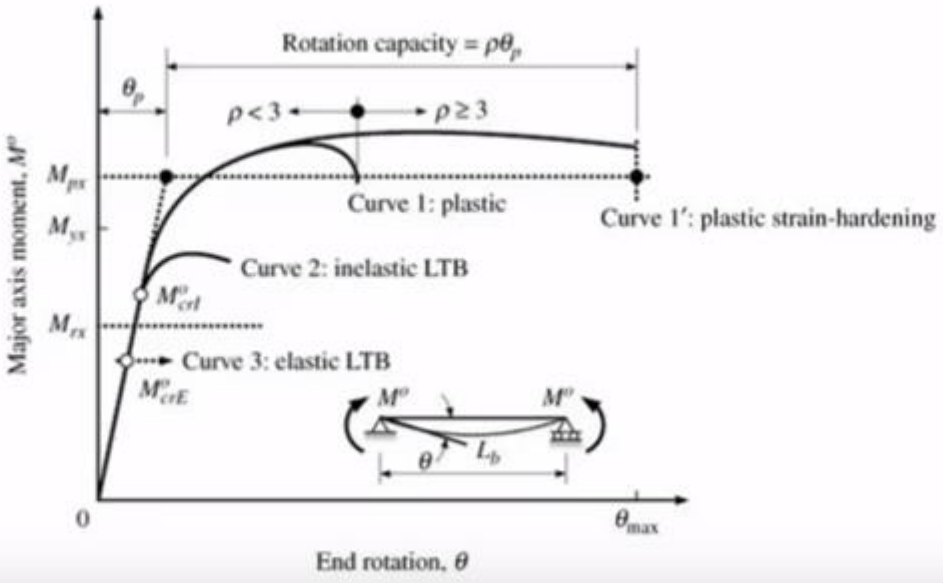
Lateral Bracing

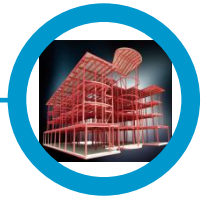
- Before we discuss capacity, recall the concept of local buckling:
  - That is, the individual plate elements might buckle before the global strength of the member is reached.
- This phenomenon is still something we have to consider for beams.
- Unlike columns, we have three ranges in which plate elements can fall:
  - Compact flanges/webs
  - Noncompact flanges/webs
  - Slender flanges/webs
- For the purposes of this course, we will limit our beams to compact flanges/webs.



Lateral Bracing

- Moment-rotation plots:
  - The main takeaway is that more compact sections can achieve more rotations (hence, more ductility)!

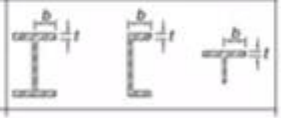




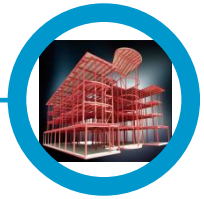
Lateral Bracing

- If we look next to the compact section limits for columns:

TABLE B4.1b
Width-to-Thickness Ratios: Compression Elements Members Subject to Flexure

Case	Description of Element	Width-to-Thickness Ratio	Limiting Width-to-Thickness Ratio: λ_p (compact/ noncompact)	Limiting Width-to-Thickness Ratio: λ_r (noncompact/ slender)	Examples
10	Flanges of rolled I-shaped sections, channels, and tees	b/t	$0.38\sqrt{\frac{E}{F_y}}$	$1.0\sqrt{\frac{E}{F_y}}$	
15	Webs of doubly symmetric I-shaped sections and channels	h/t_w	$3.76\sqrt{\frac{E}{F_y}}$	$5.70\sqrt{\frac{E}{F_y}}$	

Page-6.1-17



Continuously Braced Beam

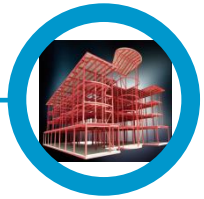
- For a beam that is continuously braced, we simply limit the beam's moments to the plastic moment capacity, M_p.
  - What we've previously defined as the ultimate load that a beam can carry.
- This is also true if there is enough discrete bracing to prevent "beam buckling".
  - Later on, we'll discuss how to determine this.

- Continuously-braced beams are governed by the following equation:

$$M_n = M_p = F_y Z_x$$
$$(\phi_b = 0.90)$$

Eqn. F2-1, Pg. 16.1-47

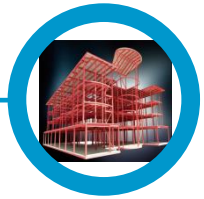
- M_p = plastic moment capacity
- F_y = yield stress (ksi)
- Z_x = plastic section modulus about the X-X axis (in^3)



Continuously Braced Beam

- The specifications provide a very useful design aid to assist in beam analysis and design.
  - *Tab. 3-2*

- This design aid will be very useful:
  - Continually-braced beam capacity
  - Discretely-braced beam capacity
  - Shear capacity



Continuously Braced Beam

Pg. 3-19 ★

- Beam design aid:
  - Both M & V capacities are listed.
- Question:
  - Why are some of the values bold?

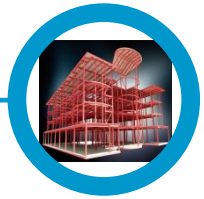
F_y = 50 ksi

Table 3-2 (continued)
W-Shapes
Selection by Z_x

Z_x

Shape	Z_x	M_{px}/Ω_b	$\phi_b M_{px}$	M_{rx}/Ω_b	$\phi_b M_{rx}$	BF/Ω_b	$\phi_b BF$	L_p	L_r	I_x	V_{nx}/Ω_v	$\phi_v V_{nx}$
		kip-ft	kip-ft	kip-ft	kip-ft	kips	kips				kips	kips
	in.^{3}	ASD	LRFD	ASD	LRFD	ASD	LRFD	ft	ft	in.^{4}	ASD	LRFD
W33×130	**467**	**1170**	**1750**	**709**	**1070**	**29.3**	**43.1**	**8.44**	**24.2**	**6710**	**384**	**576**
W27×146	464	1160	1740	723	1090	19.9	29.5	11.3	33.3	5660	332	497
W18×192	442	1100	1660	664	998	10.6	16.1	9.85	51.0	3870	392	588
W30×132	437	1090	1640	664	998	26.9	40.5	7.95	23.8	5770	373	559
W14×233	436	1090	1640	655	984	5.40	8.15	14.5	95.0	3010	342	514
W21×166	432	1080	1620	664	998	14.2	21.2	10.6	39.9	4280	338	506
W12×252[h]	428	1070	1610	617	927	4.43	6.68	11.8	114	2720	431	647
W24×146	418	1040	1570	648	974	17.0	25.8	10.6	33.7	4580	321	482

The bold section show



Continuously Braced Beam

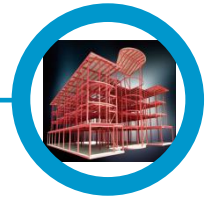
- So far, we've got moment and shear covered.
- What about deflection limits?
  - Discussed in Chapter L of the Specifications.

Chapter L: Page-6.1-163

L2. DEFLECTIONS

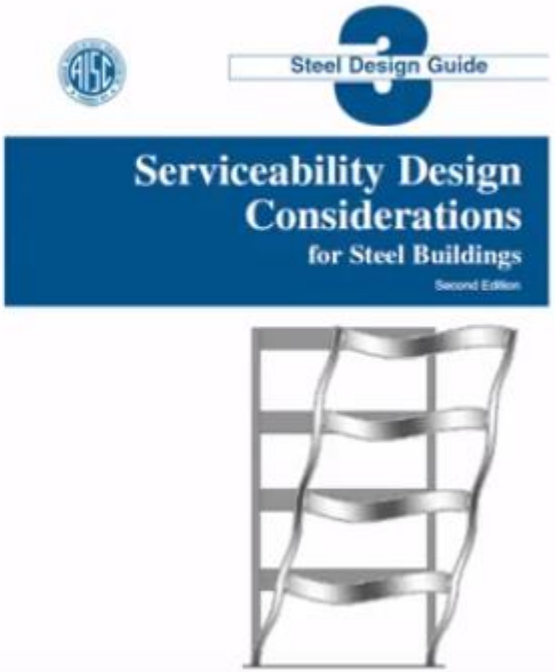
Deflections in structural members and structural systems shall be limited so as not to impair the serviceability of the structure.

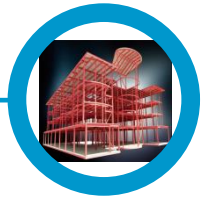
  - Therefore, we do not factor loads when assessing deflection limits.
    - We use only **service loads**!



Continuously Braced Beam

- However, unlike strength requirements such as moment and shear capacity, deflection requirements are not codified.
  - Must be agreed upon between the owner, architect, and engineer.
- There are, however, design guides which outline suggested deflection limits.
  - An example is Design Guide 3:

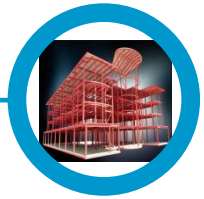




Serviceability

- Example of deflection limits from D.G. 3:

FINISH TYPE	*STRUCTURAL ELEMENT*	*DEFORMATION*	*RECOMMEN-DATION*	*LOADING*
PLASTERED CEILING	ROOF MEMBER	VERTICAL DEFLECTION	*L* / 360 MAXIMUM	0.5 × *LL* OR 50 YEAR SNOW
NON-PLASTERED CEILING	ROOF MEMBER	VERTICAL DEFLECTION	*L* / 240 MAXIMUM	0.5 × *LL* OR 50 YEAR SNOW
	FLOOR BEAM / GIRDER	VERTICAL DEFLECTION	*L* / 360 ≤ 1 IN. MAXIMUM	*DL*
PARTITION	FRAME	HORIZONTAL MOVEMENT	*H* / 500 MAXIMUM	10 YEAR WIND
	ROOF MEMBER	VERTICAL DEFLECTION	3 / 8 IN. TO 1 IN. MAXIMUM	0.5 × *LL* OR 50 YEAR SNOW
	FLOOR BEAM / GIRDER	VERTICAL DEFLECTION	*L* / 360 ≤ 3 / 8 IN. TO 1 IN., MAXIMUM	0.5 × *LL*



Serviceability

- However, these are suggested limits.
  - The owner may choose more stringent limits, based on the needs of the building.
- For the purposes of this course, if deflection limits are to be considered, they will be specified in the problem description.
  - Typically a function of live load deflection.

- We can also look up sections based on the moment of inertia.
  - *Tab. 3-3*

- These are right next to the previous beam design aid!

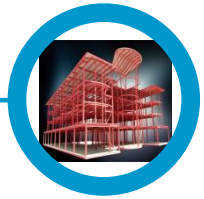
Pg. 3-28

Table 3-3 (continued)
W-Shapes
Selection by I_x

I_x

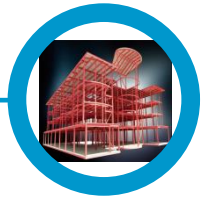
Shape	I_x in.⁴	Shape	I_x in.⁴	Shape	I_x in.⁴	Shape	I_x in.⁴
W27×84	**2850**	**W21×55**	**1140**	**W18×35**	**510**	**W12×16**	**103**
W18×143	2750	W16×77	1110	W14×48	484	W8×28	98.0
W12×252[b]	2720	W14×99	1110	W12×58	475	W10×19	96.3
W24×94	2700	W18×65	1070	W10×77	455		
W21×111	2670	W12×120	1070	W16×36	448	**W12×14**	**88.6**
W14×211	2660	W14×90	999	W14×43	428	W8×24	82.7
W18×130	2460			W12×53	425	W10×17	81.9
W21×101	2420	**W21×50**	**984**	W10×68	394	W8×21	75.3
W12×230[b]	2420	W18×60	984	W12×50	391	W10×15	68.9
W14×193	2400			W14×38	385	W8×18	61.9
		W21×48	**959**				
W24×84	**2370**	W16×67	954	**W16×31**	**375**	**W10×12**	**53.8**
W18×119	2190	W12×106	933	W12×45	348	W8×15	48.0
W14×176	2140	W18×55	890	W10×60	341	W8×13	39.6
W12×210	2140	W14×82	881	W14×34	340	**W8×10**	**30.8**

Beams



- Let's analyze a beam to see if it adequate.
  - W16×31 (A992 Steel):
    - w_D = 450 lb/ft (not including beam self-weight)
    - w_L = 550 lb/ft
    - L = 30 ft (simply supported)

- For this example:
  - We'll calculate the moment capacity.
  - We'll lookup the shear capacity (for now).
  - We'll check this beam based on a service LL deflection limit of L/360.

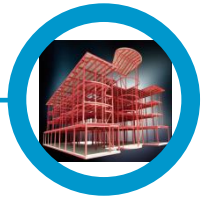


Serviceability

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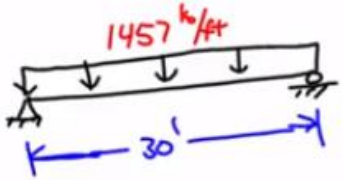
Beams



- W16x31: $Z_x = 54.0\ in^3$ $\quad b_f/2t_f = 6.28$

$I_x = 375\ in^4$ $\quad h/t_w = 51.6$

- structural analysis



$w_D = 450\ lb/ft$, $w_o = 31\ lb/ft$

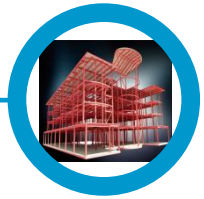
$w_L = 550\ lb/ft$

$\therefore w_u = 1.2(w_o + w_D) + 1.6 w_L$

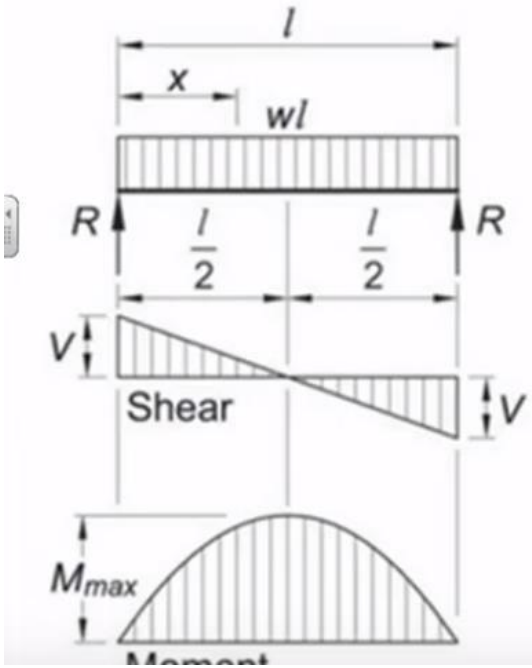
$= 1.2\,(450\ lb/ft + 31\ lb/ft) + 1.6\,(550\ lb/ft)$

$= 1457.2\ lb/ft$

Beams



1. SIMPLE BEAM — UNIFORMLY DISTRIBUTED LOAD



Total Equiv. Uniform Load ..... $= wl$

$R = V$ ..... $= \frac{wl}{2}$

V_x ..... $= w\left(\frac{l}{2} - x\right)$

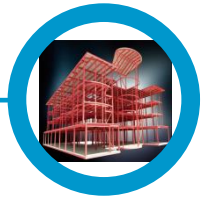
M_{max} (at center) ..... $= \frac{wl^2}{8}$

M_x ..... $= \frac{wx}{2}(l - x)$

Δ_{max} (at center) ..... $= \frac{5wl^4}{384\,EI}$

Δ_x ..... $= \frac{wx}{24\,EI}\left(l^3 - 2lx^2 + x^3\right)$

Beams



$$V_u = \frac{w_u L}{2} = \frac{(1457.2\ lb/ft)(30')}{2} = 21{,}860\ lb = 21.86\ kip$$

$$M_u = \frac{w_u L^2}{8} = \frac{(1457.2\ lb/ft)(30')^2}{8} = 163{,}900\ ft\text{-}lb = 163.9\ ft\text{-}k$$

Capacity: $\phi_b M_n = \phi_b F_y Z_x = (0.90)(50\ ksi)(54.0\ in^3) = 2430\ in\text{-}kip$

$= 202.5\ ft\text{-}kip$

Table 3-2: $\phi_b M_{px} = 203\ ft\text{-}k$

$\phi_v V_{nx} = 131\ kip$

- deflection:

$w_L = 0.55\ kip/ft$

$L = 30\ ft$

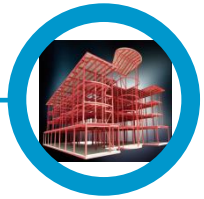
$E = 29000\ ksi$

$I_x = 375\ in^4$

$$\Delta_L = \frac{5 w_L L^4}{384 E I_x} = \frac{5(0.55\ kip/ft)(30\ ft)^4}{384(29000\ ksi)(375\ in^4)}\left(1728\ in^3/ft^3\right) = 0.922"$$

$$\Delta_{max} = \frac{L}{360} = \frac{30\ ft}{360}\left(12\ in/ft\right) = 1"$$

Beams



$\frac{b_f}{2t_f} = 6.28$ $\frac{h}{t_w} = 51.6$

Page-16.1-17

$\lambda_{pf} = 0.38\sqrt{\frac{E}{F_y}}$ $\lambda_{pw} = 3.76\sqrt{\frac{E}{F_y}}$

$= 9.15$ $= 90.55$

Beams



- Let's design a simply-supported beam:
  - The beam has continuous lateral support.
    - w_D = 400 lb/ft (not including beam self-weight)
    - w_L = 800 lb/ft
    - L = 60 ft

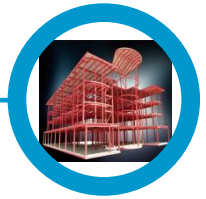
- We'll do this problem for two cases (in other words, two separate designs):
  - Case 1: No deflection limitations
  - Case 2: Live load deflection limit of L/500

Beams



- To design a continuously-braced steel beam, the procedure is pretty straightforward:
  - Determine the factored moment.
  - Determine required Z_x for a given beam *(Tab. 3-2)*.
    - This step could be skipped if you know you're using F_y = 50 ksi.
  - Look up trial section based on moment requirements.
  - Update loads (if necessary).
    - Loads may be specified to include beam self-weight.
  - Check shear and deflection requirements.
    - If deflections are likely to control a design, we can refer to our Ix selection aid.
      - *Tab. 3-3*

Beams



- analysis (moments):

$\omega_D = 400\ ^{k}/_{ft}$ * ASSUME BEAM WEIGHS 100 $^{lb}/_{ft}$ OR 0.1 $^{kip}/_{ft}$ *

$\omega_L = 800\ ^{k}/_{ft}$

$\therefore\ \omega_u = 1.2(\omega_D + \omega_D) + 1.6\,\omega_L = 1.2(0.5\ ^{kip}/_{ft}) + 1.6(0.8\ ^{kip}/_{ft}) = 1.88\ ^{kip}/_{ft}$

$$M_u = \frac{\omega_u L^2}{8} = \frac{(1.88\ ^{kip}/_{ft})(60\,ft)^2}{8} = 846\ ft\text{-}kip$$

A992 steel

$F_y = 50\ ksi$

- design #1 (no defl limits)

$M_u = 846\ ft\text{-}k = 10{,}152\ in\text{-}k$

$F_y = 50\ ksi$

$\phi M_n \geq M_u$

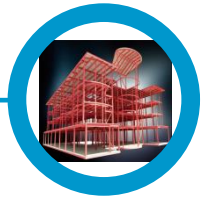
$0.9\,F_y\,Z_x \geq M_u$

$$\therefore\ Z_{x,min} = \frac{M_u}{0.9\,F_y} = \frac{10{,}152\ in\text{-}k}{0.9\,(50\ ksi)} = 225.6\ in^3$$

Try a W27x84

$Z_x = 244\ in^3$

Beams



- update load: $\omega_u = 1.2(\omega_o + \omega_D) + 1.6(\omega_L) = 1.86$ kip/ft (0.084 k/ft)

$$M_u = \frac{\omega_u L^2}{8} = 837.4 \text{ ft-k}$$

$$Z_{x_{min}} = \frac{M_u}{0.9F_y} = 223.3 \text{ in}^3$$

ANSWER

$\therefore$ Use a W24x84

$Z_x = 224 \text{ in}^3$

- design 2: $\Delta_L = \frac{5\omega_L L^4}{384EI_x}$, $\Delta_{max} = \frac{L}{500}$

$$\Delta_L \leq \Delta_{max}$$

$$\left(\frac{500}{L}\right) I_x \left(\frac{5\omega_L L^4}{384EI_x}\right) \leq \left(\frac{L}{500}\right) I_x \left(\frac{500}{L}\right)$$

$$\frac{2500\,\omega_L L^3}{384E} \leq I_x$$

$E = 29000$ ksi

$L = 60 \text{ ft} = 720''$

$\omega_L = 0.8 \text{ k/ft} = \frac{0.8}{12} \frac{\text{kip}}{\text{in}}$

Beams



$$\therefore I_{x,min} = \frac{2500\left(\frac{0.8}{12}\ k/in\right)(720")^3}{384(29000\ ksi)} = 5586.2\ in^4$$

Note

for a W24x84

$I_x = 2370\ in^4$

Try a W33x118:

$I_x = 5900\ in^4$

$Z_x = 415\ in^3$

$\phi_b M_n = 0.90 F_y Z_x$

$= 0.90(50\ ksi)(415\ in^3)$

$= 18{,}675\ in\text{-}k$

$= 1556.25\ ft\text{-}k$

$\phi V_n = 489\ kip$

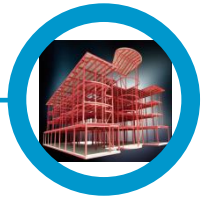
0.118 kip/ft

$w_u = 1.2(w_D + w_D) + 1.6 w_L = 1.902\ kip/ft$

$M_u = \frac{w_u L^2}{8} = 855.9\ ft\text{-}k$

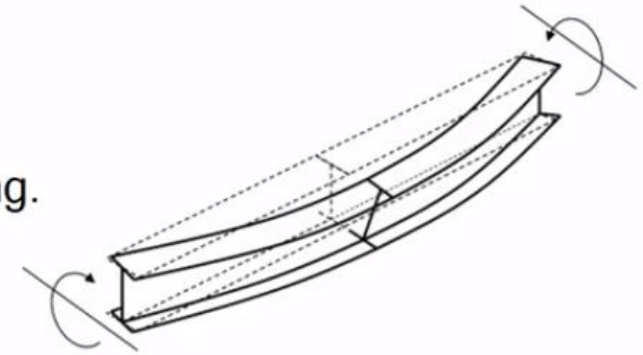
$V_u = \frac{w_u L}{2} = 57.06\ kip$

Activate Windows



Lateral Torsional Buckling

- Up until now, we haven't considered the stability issue with beams.
  - We've just assumed that the capacity will always reach M_p.
  - What if that isn't the case?
- Consider the following beam:
  - Simply supported and subjected to uniform bending.
  - The compression flange wants to buckle
  - The tension flange wants to remain straight.



- At a critical point, the beam will "kick out" and twist.
  - This phenomenon is known as lateral torsional buckling (LTB).

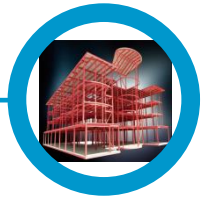
Activate Windows

Lateral Torsional Buckling

- Examples of LTB:







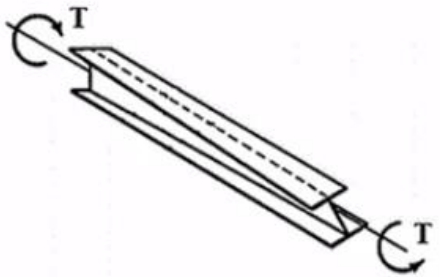
Lateral Torsional Buckling

- As stated, beams buckle by way of a torsional response.
  - Therefore, we should take some time and review torsion.

- Unlike circular shafts, I-beams respond to torsion in two fashions:

$$M_{\text{Torsion}} = M_{\text{St. Venant}} + M_{\text{Warping}}$$

  - St. Venant Torsion (or pure torsion)
  - Warping Torsion




Questions?